A Comparative Study of Periodicity and Padding in TimesNet for Multivariate Time-Series Anomaly Detection

Alberto Granati

***Abstract*—Multivariate time-series anomaly detection is difficult because normal signals can contain both local changes and repeated temporal patterns. TimesNet tries to solve this problem by discovering dominant periods using the Fast Fourier Transform (FFT) and processing them with multi-scale convolutions. This project reconstructs TimesNet for unsupervised anomaly detection on Server Machine Dataset (SMD) and studies which aspect of this architecture gives a benefit impact. The original TimesNet is compared with three other architectures: a non periodic 1D ablation, a periodic 1D ablation, and a fractional period 2D block inspired by a recent extension of TimesNet. All variants use the same data splits, training procedure, reconstruction loss, and evaluation protocol. Across seven random seeds, the 2D variants achieve the highest adjusted F1-scores, but the improvement over the non-periodic 1D baseline is limited. Raw point-wise F1-scores remain similar across all models, while the 2D architectures require substantially more parameters and training time. These results suggest that the periodic 2D representation provides a modest benefit under event-level evaluation, but with a considerable computational cost.**

***Index Terms*—anomaly detection, multivariate time series, TimesNet, TimesBlock, reconstruction error, periodicity**

I. Introduction

Time-Series anomaly detection aims to identify unexpected behaviour of system during time. This task is important for service monitoring, predictive maintenance and fault diagnosis. It's also challenging: a variation in a specific context might be a real anomaly or a normal value in others contexts. Usually in unsupervised settings, the model is trained on normal data, trying to learn the correct behaviour of the system, and then it's tested data that has anomalies inside, without seeing the labels.

TimesNet was proposed as a general architecture to solve sevelar time series task such as forecasting and anomaly detection [1]. Its main innovation is to represent the temporal variation in two dimensions, using its main frequencies by FFT. Selecting the highest frequencies are used in order to reorganised data in periods in order to build a matrix form and then apply a 2D convolution. In this way the 2D convolution can benefit from both intra period information and inter period information.

This projects studies TimesNet trying to figure out which are the real aspect that give TimesNet the capability to obtain the best result in terms of anomaly detection scores. Moreover we will have a look to a proposal done by [2], in which some limitations of TimesNet are been solved. Then, this work reconstructs a TimesNet reconstruction model and makes 3 controlled variants available, while keeping the remaining pipeline fixed.

- a non-periodic 1D ablation that removes the FFT and reshape, but preserves the multi-kernel structure through parallel 1D convolutions;
- a periodic 1D ablation that uses FFT to catch the periodicity of the time series and then use it to catch inter-period relation using 1D convolutions;
- a fractional-period TimesBlock that keeps decimal period lengths and pads individual segments, following the methods proposed by Nagahama et al. [2].

Of course, there is a comparison with the standard periodic TimesBlock with FFT-based period selection and 2D Inception-style convolutions;

Now let's have a look to the structure of the paper. Section II gives the theoretical background. Section III presents the complete pipeline, while Section IV describes the dataset and preprocessing. The model variants and evaluation rules are detailed in Section V. Results, limitations, and conclusions are reported in Section VI and Section VII.

II. Related Work

A. Reconstruction-Based Anomaly Detection

The Server Machine Dataset (SMD) was introduced with OmniAnomaly for multivariate monitoring data [3]. A common unsupervised approach trains a reconstruction model on normal observations. Given an input $\boldsymbol{x}_t$ and its reconstruction $\hat{\boldsymbol{x}}_t$, a large reconstruction error is interpreted as evidence of an anomaly. This approach does not require anomaly labels during training, but its result depends on model capacity and on the rule used to convert a continuous score into a binary prediction.

An important evaluation detail is point adjustment. The procedure used in the notebook follows the event-level convention also used in anomaly-detection benchmarks related to Anomaly Transformer [4]. If one point of a ground-truth anomaly interval is detected, all points in that interval are changed to positive predictions. This can be useful for event detection, but the adjusted result is different from strict point-wise detection: a single true positive can yield credit for an entire, potentially long, anomaly interval and can therefore substantially increase recall and F1-score.

B. TimesNet and Fractional Periods

TimesNet models multiple periodicities instead of processing only adjacent 1D samples [1]. For each TimesBlock, the FFT amplitudes are averaged over the batch and channels. The k strongest non-zero frequencies define candidate periods. Each representation is processed by a parameter-efficient Inception block with several 2D kernel sizes, inspired by the multi-scale design of GoogLeNet [5]. The results are converted back to 1D and aggregated using softmax-normalised FFT amplitudes. Their analysis is done only with respect to other arcihtectures. My contribution is to check in more details which are the strength of this approach analyzing both the periodicity and the use of 2D convolution kernels.

Nagahama et al. observed that integer period lengths may misalign equal phases when T/f is not an integer [2]. Their first method keeps T/f as a decimal value, rounds each segment boundary, and zero-pads shorter segments to a common width. Their second method fills missing positions with a signal value instead of zero. Their results show that the model performs better, but this is not done in a classic anomaly detection problem and their analysis is not done with respect to the SMD dataset.

III. Processing Pipeline

The overall processing pipeline is shared by all four architectures considered in this work. The models differ only in the internal structure of the temporal processing block, allowing us to isolate the contribution of periodicity and of the two-dimensional representation introduced by TimesNet.

Given an input window of the multivariate time series, the data are first normalized and projected into a latent representation through an embedding layer. The embedded sequence is then processed by a stack of TimesNet blocks. Depending on the architecture under consideration, this stage uses one of four different temporal processing strategies described below. After each block, layer normalization is applied. Finally, the latent representation is projected back to the original feature space and de-normalized, producing a reconstruction of the input sequence.

Since the models are trained for reconstruction-based anomaly detection, normal observations are expected to be reconstructed accurately, whereas anomalous observations should produce a larger reconstruction error. The reconstruction error is therefore used as the anomaly score and compared with a threshold to identify anomalous samples.

The first architecture is the **original TimesNet**, which follows the structure proposed in [1]. TimesNet first uses the Fast Fourier Transform (FFT) to identify the dominant periodicities of the input signal. For each selected period, the one-dimensional sequence is reorganized into a two-dimensional representation whose dimensions describe variations within a period and across successive periods. Multi-scale 2D convolutions are then applied to this representation, and the outputs associated with the different detected periods are adaptively combined according to their FFT amplitudes.

The **non-periodic 1D ablation** removes both the FFT-based period detection and the two-dimensional transformation. The embedded sequence is processed directly along the temporal dimension using multiple parallel 1D convolutions with different kernel sizes. Their outputs are averaged, following the multi-scale idea of the Inception block used by TimesNet. This architecture therefore retains multi-scale temporal feature extraction while removing any explicit modelling of periodicity.

The **periodic 1D architecture** retains the FFT-based period detection but avoids the two-dimensional representation. Two complementary 1D processing branches are used. A local branch applies the same multi-scale convolutions used in the non-periodic ablation and captures short and medium range temporal dependencies. In parallel, an inter-period branch uses each period detected by the FFT as the dilation of the convolutional kernels. In this way, the convolution can directly relate observations located at corresponding positions in consecutive cycles. The local and inter-period representations are combined for each detected period, after which the different period-specific outputs are adaptively aggregated using the FFT amplitudes. This variant allows us to investigate whether explicitly modelling periodic relationships is beneficial even without constructing a 2D representation.

Finally, the **modified 2D TimesNet** preserves the main processing mechanism of the original TimesNet, including FFT-based period detection and 2D convolutions, while modifying the construction of the periodic representation. In the original model, periods are approximated by an integer length and any required padding is appended to the end of the complete sequence before reshaping. In the modified architecture, the period obtained from the FFT is kept as a fractional value when defining the boundaries of successive periodic segments. Each segment is then padded individually to a common width using its last observed value. The resulting segments can consequently be arranged into a regular 2D representation while more closely preserving the periodic structure identified by the FFT.

The four architectures can therefore be interpreted as progressively introducing different components of the TimesNet mechanism: multi-scale temporal processing, explicit periodic information, and two-dimensional intra- and inter-period modelling. Since the remaining reconstruction pipeline is kept unchanged, their comparison provides a controlled analysis of the contribution of these architectural choices to anomaly detection performance.

IV. Signals and Features

The experiments are conducted on the Server Machine Dataset (SMD), represented as multivariate time series with 38 channels. The local training array contains $708{,}405$ temporal samples, while the test array contains $708{,}420$ samples. Each temporal sample is therefore represented by a 38-dimensional vector. The test set is associated with one binary label for each time point, indicating whether the corresponding observation

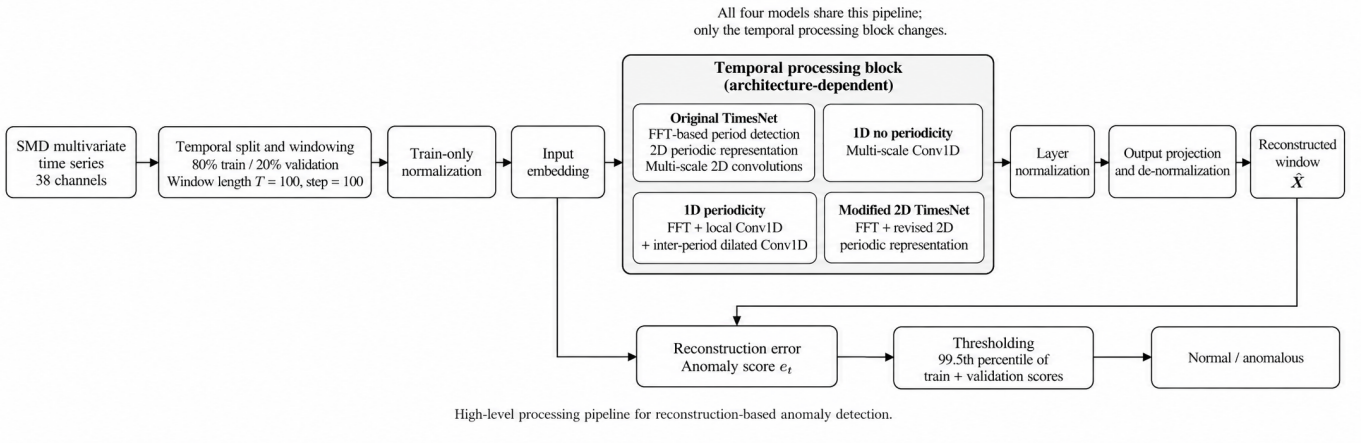


Fig. 1. High-level processing pipeline for reconstruction-based anomaly detection. All four architectures share the same overall pipeline and differ only in the temporal processing block.

is normal or anomalous. In total, the test sequence contains $29{,}444$ anomalous points.

Before constructing the input sequences, the original normal training series is divided chronologically into training and validation subsets. The first 80%, is used for training, while the remaining 20%, is used for validation. The original test sequence is kept unchanged and is used only for the final evaluation. This temporal split preserves the ordering of the observations and prevents information from later portions of the normal sequence from entering the training subset.

The signals are standardised independently for each channel. The scaler is fitted exclusively on the training subset, and the resulting mean and standard deviation are then applied to the validation and test sequences. In this way, no information from the validation or test sets is used to estimate the normalisation parameters, avoiding data leakage. No additional filtering, denoising, interpolation, or missing-value reconstruction is applied in the implemented pipeline.

After standardisation, each sequence is divided into fixed-length temporal windows. Each input sample has shape

$$X \in \mathbb{R}^{100\times 38},$$

where 100 is the number of consecutive temporal samples and 38 is the number of measured channels. The window length is set to $T = 100$ and the step is also set to 100, so consecutive windows do not overlap. Only complete windows are retained; any residual samples at the end of a sequence that are insufficient to form a full window are discarded.

No handcrafted feature vector is extracted from each window. Instead, the complete 100×38 multivariate segment is directly provided to the neural network. The model subsequently projects the 38 input variables into a latent representation through its embedding layer, while the following temporal processing blocks learn the relevant features automatically. Therefore, in this work the feature extraction stage is performed by the network itself rather than through an external preprocessing procedure.

During training, the order of the training windows is shuffled by the data loader, whereas validation and test windows are processed without shuffling so that their temporal ordering is preserved. The labels associated with the test sequence are not used to train the reconstruction model, but only during the final anomaly-detection evaluation.

V. Learning Framework

The four architectures are trained within the same reconstruction-based learning framework. Each input window is first mapped to a latent representation and then processed by three temporal blocks. The specific temporal block depends on the architecture under consideration, while the remaining model structure, training objective, optimizer, learning-rate schedule, and evaluation procedure are kept unchanged. This allows the comparison to focus on the effect of the different temporal representations rather than on differences in the training procedure.

A. Temporal Processing Blocks

The standard 2D TimesNet block first identifies the dominant periodic components of the latent sequence through the Fast Fourier Transform. Given an input representation $\boldsymbol{X}$, the amplitude associated with frequency index f is computed as

$$A(f) = \mathrm{Avg}_{b,c}\left|\mathrm{FFT}(\boldsymbol{X})_{b,f,c}\right|. \tag{1}$$

The zero-frequency component is discarded and the $k = 3$ dominant frequencies are selected. For each selected frequency f_i, the standard TimesNet block defines the corresponding integer period as

$$p_i = \left\lfloor \frac{T}{f_i} \right\rfloor. \tag{2}$$

The temporal sequence is then padded at its end so that its length becomes divisible by p_i, and the representation is rearranged from a one-dimensional temporal sequence into a two-dimensional periodic representation. This representation

is processed by two multi-scale 2D Inception blocks separated by a GELU activation. Each Inception block applies square convolutional kernels of sizes $1, 3, 5, 7, 9$, and 11, and averages the resulting feature maps.

The output corresponding to each selected period is converted back to the temporal domain. The different period-specific representations are then combined according to their FFT amplitudes. After applying a softmax to the period weights, the final block output is

$$\boldsymbol{Y} = \boldsymbol{X} + \sum_{i=1}^{k} w_i \boldsymbol{Y}_i, \tag{3}$$

where w_i is the normalized importance of the i-th detected period and $\boldsymbol{Y}_i$ is the corresponding processed representation. The residual connection preserves the original latent sequence.

The **1D non-periodic variant** removes the FFT-based period detection, the two-dimensional reshaping, and the adaptive period aggregation. Instead, the latent sequence is processed directly through parallel one-dimensional convolutions with kernel lengths $1, 3, 5, 7, 9$, and 11. Their outputs are averaged and passed through the same two-block structure with GELU and a residual connection. This variant therefore preserves multi-scale temporal processing while removing explicit periodic modelling.

The **1D periodic variant** retains the FFT-based identification of the $k = 3$ dominant periods but does not construct a two-dimensional representation. It combines two one-dimensional branches. The first is a local multi-scale branch identical to the non-periodic 1D model. The second is an inter-period branch in which the detected period is used as the dilation factor of the convolution. For each detected period, the outputs of the local and inter-period branches are averaged, and the resulting period-specific representations are adaptively combined using the same FFT-based weights as in the standard TimesNet block.

The **modified 2D TimesNet** keeps the FFT-based period selection and two-dimensional convolutional processing but modifies the construction of the periodic representation. Instead of approximating the period directly by an integer value, the real-valued period is retained:

$$p_i = \frac{T}{f_i}. \tag{4}$$

For periodic segment j, the segment boundaries are determined as

$$h_j = \lfloor j p_i + 0.5 \rfloor, \qquad e_j = \lfloor (j+1) p_i + 0.5 \rfloor. \tag{5}$$

The resulting segments may have slightly different lengths. Each segment is therefore padded to a common width $\lceil p_i \rceil$ by repeating its last observed value. The padded segments are stacked into a regular two-dimensional representation and processed by the same 2D Inception module used by the original TimesNet. After convolution, the padding introduced in each segment is removed and the temporal sequence is reconstructed.

TABLE I
IMPLEMENTED TEMPORAL PROCESSING VARIANTS.

Variant	Periodicity	Processing	Parameters
1D non-periodic	None	1D multi-scale	896,934
1D periodic	FFT-based	1D local + dilated	1,119,270
2D TimesNet	$\lfloor T/f \rfloor$	2D multi-scale	7,040,934
Modified 2D	T/f	2D fractional	7,040,934

Table I summarizes the main controlled differences among the four architectures.

B. Training Strategy

All four architectures are trained as reconstruction models using only the normal training sequence. Given an input window $\boldsymbol{X}$, the network produces a reconstruction $\hat{\boldsymbol{X}}$, and training minimizes the mean squared reconstruction error

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{TC} \sum_{t=1}^{T} \sum_{c=1}^{C} \left(x_{t,c} - \hat{x}_{t,c} \right)^2. \tag{6}$$

The model parameters are optimized using Adam with an initial learning rate of 10^{-4}. An exponential learning-rate scheduler is applied after each epoch with decay factor gamma=0.5.

Training is performed for at most 10 epochs. After each epoch, the reconstruction loss is evaluated on the validation split. Whenever the validation loss decreases, the current model parameters are stored as the best checkpoint. Early stopping is triggered if the validation loss does not improve for three consecutive epochs. At the end of training, the best validation checkpoint is restored and used for the final evaluation.

C. Anomaly Scoring and Decision Rule

After training, anomaly detection is performed using the reconstruction error. For each time step t, the squared reconstruction errors are averaged across the $C = 38$ channels:

$$E_t = \frac{1}{C} \sum_{c=1}^{C} \left(x_{t,c} - \hat{x}_{t,c} \right)^2. \tag{7}$$

The anomaly energies computed on the training and validation sequences are concatenated to form the reference distribution of normal reconstruction errors. With an anomaly ratio of $r = 0.5\%$, the detection threshold is selected as

$$\tau = Q_{99.5} \left(\{ E_t^{\mathrm{train}} \} \cup \{ E_t^{\mathrm{val}} \} \right), \tag{8}$$

where $Q_{99.5}$ denotes the 99.5th percentile of the combined reconstruction-error distribution. A test point is classified as anomalous when

$$\hat{y}_t = [E_t > \tau]. \tag{9}$$

The ground-truth test labels are therefore not used to determine the reconstruction model or the anomaly threshold.

D. Evaluation Procedure

Performance is first evaluated using the raw point-wise predictions. Accuracy, precision, recall, and F1-score are computed from the predicted and ground-truth labels.

Then, I compute the adjusted metrics using the event-level adjustment procedure commonly adopted for SMD experiments. If at least one predicted anomalous point falls inside a contiguous ground-truth anomaly interval, the entire corresponding interval is marked as anomalous. Predictions outside ground-truth anomalous intervals remain unchanged. Raw and adjusted metrics are reported separately in order to distinguish strict point-wise detection performance from event-level detection performance.

VI. Results

The four architectures were evaluated using the same training, validation, and test partitions, as well as the same reconstruction-based anomaly-detection procedure. Each model was trained using seven different random seeds in order to reduce the influence of random initialization. The reported values correspond to the mean and standard deviation across these runs.

A. Overall Performance

Table II reports the main results for the four architectures. Both raw point-wise F1 and adjusted event-level metrics are reported. The number of trainable parameters and the average training time are also included to highlight the computational cost associated with each architectural choice.

The raw point-wise F1 scores are low across all architectures, ranging from 0.1059 to 0.1098. Figure 2 shows the considerable difference between raw and adjusted F1. This confirms that the event-level adjustment has a strong influence on the apparent performance and that adjusted results should be interpreted carefully.

A small benefit emerges when considering adjusted F1. The non-periodic 1D model obtains an adjusted F1 of 0.8242 ± 0.0050, whereas the modified 2D model reaches 0.8290 ± 0.0056. The difference between the worst and best model is therefore only 0.0048. Since F1 is the harmonic mean of precision and recall, this result indicates a modest improvement rather than a substantial difference in anomaly-detection performance.

B. Effect of Periodicity and Dimensionality

The comparison between the two 1D variants suggests that introducing inter-period information can improve the adjusted result. The period-aware 1D model increases adjusted F1 from 0.8242 to 0.8268. This may indicate that FFT-based period detection and dilated inter-period convolutions capture useful periodic dependencies.

However, this interpretation is not unambiguous. The period-aware 1D model combines local and inter-period convolutional branches and consequently contains more parameters than the non-periodic 1D model. The observed improvement may therefore be caused both by the explicit modelling of periodic dependencies and by the increased capacity of the model. Conversely, a larger number of parameters does not necessarily guarantee better performance and may increase the risk of overfitting.

The two-dimensional variants obtain the highest adjusted F1 scores. Standard 2D TimesNet reaches 0.8287 ± 0.0053, while the modified fractional-padding version reaches 0.8290 ± 0.0056. Although this result suggests a small advantage for the periodic 2D representation, the differences remain limited relative to the variability across seeds. Figure 2 illustrates this behaviour: the relative ordering of the models changes across seeds, while all adjusted F1 values remain within a narrow range. The modified 2D architecture was inspired by the fractional-period and repeated-value padding strategy proposed in *Improvement of Anomaly Detection Through Enhanced Feature Extraction in TimesNet*. On the SMD dataset, this modification does not provide a meaningful improvement in anomaly-detection performance, but its average training time is lower, requiring 135.30 seconds instead of 143.99 seconds. This observation suggests that constructing periods through fractional segmentation and repeated-value padding may provide a small computational advantage in this experimental setting. However, further experiments would be needed to establish whether the observed speed-up is consistently caused by the padding strategy itself.

C. Qualitative Analysis

Finally, Figure 3 presents a qualitative comparison of reconstruction energies and false positives over a selected portion of the test sequence. The figure illustrates that the two models produce different energy distributions and thresholds. These examples are useful for understanding the behaviour of the models on individual anomaly intervals, but they should be interpreted as a case study rather than as general evidence of superior performance.

Overall, the experiments suggest that the periodic 2D TimesNet representation provides a modest benefit under the adjusted metric, but this benefit comes with a considerable computational cost. Adding periodic information to the 1D model also slightly improves the adjusted F1-score, although this effect is confounded by the increased number of parameters. The modified fractional-padding strategy does not improve anomaly-detection performance on SMD, but it appears to reduce the training time of the 2D model.

VII. Conclusion

This work compared the original 2D TimesNet architecture with three variants designed to study the role of periodicity and two-dimensional temporal modelling in reconstruction-based anomaly detection on SMD.

The experiments show that the periodic 2D variants achieve the best adjusted F1-scores, although the improvement over the 1D models is small and does not produce a clear advantage under the raw point-wise metric. Adding inter-period information to the 1D model provides only a limited gain, while the fractional-padding modification does not improve detection

TABLE II
PERFORMANCE COMPARISON ACROSS 7 RANDOM SEEDS. VALUES ARE REPORTED AS MEAN ± STANDARD DEVIATION.

Model	Parameters	Raw F1	Adjusted Precision	Adjusted Recall	Adjusted F1	Training time [s]
1D non-periodic	896,934	0.1059 ± 0.0061	0.8188 ± 0.0111	0.8300 ± 0.0153	0.8242 ± 0.0050	19.51 ± 0.25
1D periodic	1,119,270	0.1080 ± 0.0051	0.8139 ± 0.0087	0.8401 ± 0.0064	0.8268 ± 0.0063	34.88 ± 0.47
2D TimesNet	7,040,934	0.1095 ± 0.0023	0.8196 ± 0.0068	0.8381 ± 0.0061	0.8287 ± 0.0053	143.99 ± 0.74
Modified 2D	7,040,934	0.1098 ± 0.0023	0.8201 ± 0.0071	0.8381 ± 0.0061	0.8290 ± 0.0056	135.30 ± 1.75

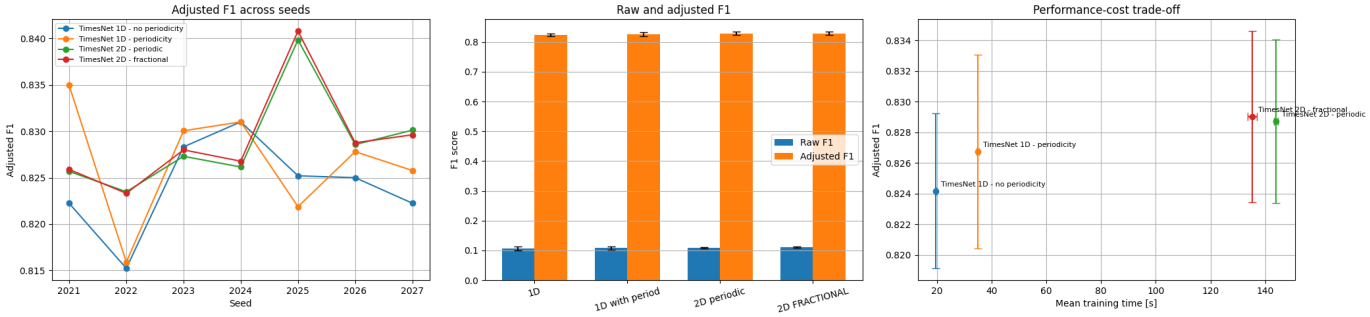


Fig. 2. In the leftmost graph is shown the adjusted F1 per each seed of each model. In the center we have the difference between the raw F1 and the adjested F1. In the right the we show the F1 related to the training time.

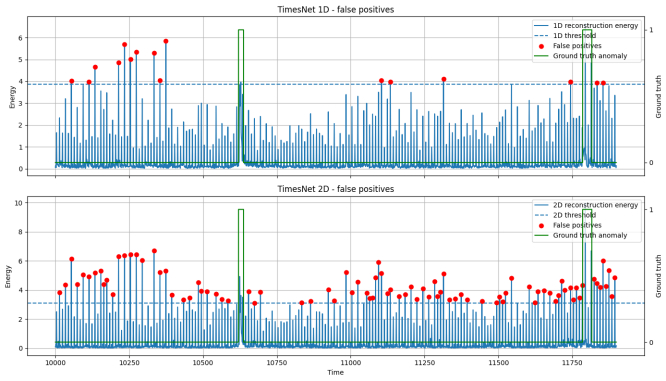


Fig. 3. When the energy goes up to the threshold, then the model detects an anomaly.

performance over standard TimesNet. It does, however, reduce the observed training time.

Overall, the results suggest that the 2D periodic representation provides a modest benefit, but at a substantially higher computational cost. Future work should verify these conclusions on additional datasets and under different period-selection and window-length settings.

At the beginning of the project, choosing a suitable topic and defining meaningful model variants was not so easy, because I was not sure what to do. However, studying the TimesNet paper and its reference implementation made it possible to understand the architecture and its main design choices. I was also surprised that the differences among the tested architectures were smaller than expected, since explicitly modelling periodicity and two-dimensional temporal variations produced only modest improvements. Moreover, the standard TimesNet model achieved a lower adjusted F1-score than the value reported in the original paper (0.8287 compared with 0.8512); the reason is written here because it's something that I notice only when the report has already been written completely. In the implementation of the Time-Series Library, I noticed that the threshold was computed using also the test set, while in my approach only the training (and validation) data were used. This might be one possible reason.

REFERENCES

[1] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "TimesNet: Temporal 2d-variation modeling for general time series analysis," in *The Eleventh International Conference on Learning Representations*, 2023.

[2] H. Nagahama, M. Yoshioka, K. Inoue, M. Todorokihara, and K. Omori, "Improvement of anomaly detection through enhanced feature extraction in TimesNet," in *Distributed Computing and Artificial Intelligence, 21st International Conference*, vol. 1259 of *Lecture Notes in Networks and Systems*, pp. 11–20, Springer Nature Switzerland, 2025.

[3] Y. Su, Y. Zhao, C. Niu, R. Liu, W. Sun, and D. Pei, "Robust anomaly detection for multivariate time series through stochastic recurrent neural network," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 2828–2837, Association for Computing Machinery, 2019.

[4] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations*, 2022.

[5] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *2015 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1–9, 2015.